AEGIS: Mining Graph Structure for Adversarial Vulnerability Analysis of Graph Neural Networks

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***Abstract*—We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a structural vulnerability analysis framework for graph neural networks. AEGIS constructs a constrained sensitivity matrix S_c that maps symmetric, edge-only adjacency perturbations to hidden-state shifts, yielding SVD-optimal attacks, per-edge vulnerability rankings, and per-node first-order sensitivity radii. The S_c computation applies to any differentiable GNN: for K-layer models, it is derived from the unrolled Jacobian $\partial Z_K/\partial A$; for contractive implicit GNNs (IGNN-class), it is derived from the equilibrium sensitivity $(I - J_z)^{-1}J_A$, which additionally provides a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\left\|W\right\|_2$ and convergence guarantees not available for explicit architectures. Under constrained perturbations, first-order tightness is 1.00 ± 0.01 at $\varepsilon = 0.01$ and remains within 15% at $\varepsilon = 0.10$ across 7 architectures (IGNN, GCN, GIN, GAT, GraphSAGE, APPNP) and 9 datasets (10 seeds each). The SVD-optimal attack inflicts 2–8× more damage than edge-constrained random perturbation. As a case study, the vulnerability spectrum recovers power grid N-1 contingency rankings with precision@10 between 0.66 and 0.81. Code will be released upon publication.**

***Index Terms*—Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem**

I. Introduction

Graph neural networks are vulnerable to adversarial structural perturbation: small changes to graph topology can drastically alter predictions [1], [2], [3], [4]. Defending against such attacks requires understanding which perturbations matter and by how much. Empirical defenses such as robust aggregation [5], [6] and graph structure learning [7], [8] improve resilience but lack formal guarantees. Certified approaches either provide probabilistic bounds via randomized smoothing [9], [10], require layer-by-layer propagation tied to specific architectures [11], [12], [13], or target explicit models exclusively [14], [15]. None provides a unified structural vulnerability analysis that works across GNN families.

We introduce AEGIS, a framework built on a single object: the *constrained sensitivity matrix* S_c, which maps symmetric, edge-only perturbations of the adjacency matrix to hidden-state shifts. The S_c computation applies to any differentiable GNN: for a K-layer model, S_c is derived from the unrolled Jacobian $\partial Z_K/\partial A$ (theorem 4); for contractive implicit GNNs (IGNN-class [16]), S_c is derived from the equilibrium sensitivity $S = (I-J_z)^{-1}J_A$ via the implicit function theorem, which additionally provides a critical perturbation budget $\varepsilon_{\text{crit}}$ and convergence guarantees (theorem 1) not available for explicit architectures. In both cases, S_c yields: (i) an SVD-optimal structural attack, (ii) a per-edge vulnerability ranking, and (iii) per-node first-order sensitivity radii (locally tight but not global certificates without second-order bounds).

The central finding is that constrained first-order predictions via S_c achieve tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ and remain within 15% at $\varepsilon = 0.10$ across 7 GNN architectures (IGNN, GCN, GIN, GAT, GraphSAGE, APPNP) and 9 datasets spanning 4 domains.

Contributions.

1) **Constrained sensitivity framework.** We derive the constrained sensitivity matrix S_c for any differentiable GNN (theorem 4). For contractive implicit models, we additionally prove a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\left\|W\right\|_2$ and three vulnerability regimes (theorem 1) that characterize when first-order guarantees hold.
2) **Unified vulnerability analysis.** From S_c we extract SVD-optimal attacks (2–8× stronger than random across all architectures), per-edge vulnerability spectra, and per-node first-order sensitivity radii. We validate on 7 architectures: IGNN [16], GCN [17], GIN, GAT, GraphSAGE, and APPNP (section V-H).
3) **Cross-domain validation.** We validate across 9 datasets in 4 domains: citation networks [18], e-commerce [19], encyclopedia [20], and IEEE power grids [21]. The vulnerability spectrum recovers power grid N-1 contingency rankings with precision@10 = 0.66–0.81 (section VI).

II. Preliminaries and Threat Model

This section establishes the notation, formalizes the model class under consideration, specifies the threat model, and states the problem that AEGIS addresses.

A. Notation and Formalization

Let $G = (V, E, A)$ denote an undirected graph with $N = |V|$ nodes, edge set E, and adjacency matrix $A \in \mathbb{R}^{N\times N}$. Each node $v \in V$ carries a feature vector $x_v \in \mathbb{R}^{d_{\text{in}}}$, collected into $X \in \mathbb{R}^{N\times d_{\text{in}}}$. We write $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ for the symmetric normalized adjacency with self-loops, where D is the degree matrix of $A + I$.

Deep equilibrium GNNs. A DEQ-GNN [22], [23] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it from $Z_0 = 0$ until convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad z^* = \lim_{k\to\infty} Z_k. \tag{1}$$

When F_θ is contractive in operator norm ($\kappa = \left\|J_z\right\|_2 < 1$, where $J_z = \partial F/\partial z|_{z^*}$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$ regardless of initialization.

IGNN instantiation. The Implicit Graph Neural Network (IGNN) [16] uses the operator

$$F(Z, A) = \sigma(\hat{A} Z W^\top + X_{\text{proj}}), \tag{2}$$

where $W \in \mathbb{R}^{d\times d}$ is the shared weight matrix, $X_{\text{proj}} \in \mathbb{R}^{N\times d}$ is a learned feature projection, and σ is ReLU activation. Contractivity is enforced by constraining $\|W\|_2$ via spectral normalization [24], ensuring $\kappa \leq \left\|\hat{A}\right\|_2 \cdot \|W\|_2 < 1$. Training uses implicit differentiation through the fixed point [25], [26], and Jacobian regularization [27] can further stabilize convergence. A linear classification head $f(z^*) = W_{\text{cls}} z^* + b$ maps the equilibrium to class predictions.

Implicit function theorem. At the equilibrium, the equation $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A. The IFT gives the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{3}$$

where the resolvent $(I - J_z)^{-1}$ exists and satisfies $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ by the Neumann series (convergent because $\kappa < 1$). This bound uses the operator norm $\kappa = \|J_z\|_2$; the weaker spectral-radius bound $1/(1-\rho)$ holds only for normal J_z. The pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho) \geq 1$ [28] quantifies this gap, with $\eta = 1$ for normal matrices. The IFT has been used for hyperparameter optimization [29] and implicit differentiation [25], but not for adversarial structural analysis.

B. Threat Model

We consider a white-box attacker who has full access to the trained IGNN model and perturbs the normalized adjacency matrix:

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \text{subject to} \quad \left\|\delta\hat{A}\right\|_F \leq \varepsilon. \tag{4}$$

The perturbation is further constrained to be:

- **Symmetric:** $\delta\hat{A} = \delta\hat{A}^\top$ (preserving the undirected nature of the graph).
- **Edge-only:** $[\delta\hat{A}]_{ij} = 0$ for $(i,j) \notin E$ (only existing edge weights may be modified; no new edges are introduced).
- **Continuous:** edge weights are perturbed continuously in $\mathbb{R}$, not discretely flipped.

This threat model captures continuous edge-weight perturbations to the normalized message-passing operator. It aligns naturally with the IGNN architecture, where $\hat{A}$ appears directly in F (eq. (2)). Discrete edge insertions or deletions, which change the graph topology and require recomputing the degree normalization $D^{-1/2}$, are outside the formal guarantee.

The key distinction from input-feature perturbation (studied for DEQs in [30], [31]) is that structural perturbation changes the *operator itself*: the equilibrium depends on A through both the state Jacobian $J_z = \text{diag}(\sigma') \cdot (\hat{A} \otimes W)$ and the adjacency Jacobian $J_A = \partial F / \partial \text{vec}(A)$. This dual dependence makes structural sensitivity qualitatively different from input sensitivity.

C. Problem Statement

Given a trained contractive implicit GNN with equilibrium z^* on graph G, we aim to answer three questions:

1) **Vulnerability ranking:** Which edges, if perturbed, cause the largest shift in the equilibrium? That is, find a ranking of edges $(i,j) \in E$ by their adversarial impact on z^*.
2) **Optimal attack:** What is the worst-case perturbation $\delta\hat{A}^*$ that maximizes $\|\Delta z^*\|_F$ subject to the threat model constraints in eq. (4)?
3) **Robustness assessment:** For each node v, what is the largest perturbation budget r_v such that any $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of v?

AEGIS addresses all three questions through a unified object: the constrained sensitivity matrix S_c derived from the IFT applied to the equilibrium equation. The following sections formalize this construction (section III) and describe the computational pipeline (section IV).

III. Adversarial Equilibrium Theory

Having formalized the problem in section II, we now derive the theoretical foundation for answering the three questions posed there. The main result (theorem 1) characterizes vulnerability regimes as a function of perturbation magnitude, while the constrained sensitivity matrix S_c provides the unified object from which vulnerability rankings, optimal attacks, and robustness radii are all extracted.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, A) = \sigma(AZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

(A1) *σ is ReLU (or any 1-Lipschitz activation with $\sup|\sigma'| = 1$);*

(A2) *W is spectral-norm constrained: $\|W\|_2 \leq c$;*

(A3) *The state Jacobian satisfies $\|J_z\|_2 \leq \kappa < 1$ at the fixed point z^* (contraction in operator norm).*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{5}$$

Then structural perturbations δA with $\|\delta A\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{6}$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1-\kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\text{crit}}$): The resolvent norm $\left\|(I - J_z')^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$. Intuitively, as the perturbation approaches the critical budget, the system's sensitivity to further perturbation grows without bound.*

(c) Supercritical *($\varepsilon > \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J_z'\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation* *direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and the first-order guarantees from part (a) cease to apply.*

The intuition behind this three-regime picture is physical: a contractive system has a restoring force that pulls trajectories toward the equilibrium. Small perturbations shift the equilibrium smoothly (subcritical). As the perturbation approaches $\varepsilon_{\mathrm{crit}}$, the restoring force weakens until the system becomes critically sensitive. Beyond $\varepsilon_{\mathrm{crit}}$, the restoring force may vanish entirely, and the equilibrium can bifurcate or disappear.

Proof. **Part (a).** The Jacobian of F with respect to $\mathrm{vec}(Z)$ takes the form $J_z = \mathrm{diag}(\sigma') \cdot (\hat{A} \otimes W)$, where $\sigma' \in \{0,1\}^{Nd}$ for ReLU (A1). The operator norm satisfies $\kappa = \|J_z\|_2 \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 \cdot \sup|\sigma'| = \left\|\hat{A}\right\|_2 \cdot \|W\|_2$. Applying the IFT to $G(z,A) = z - F(z,A) = 0$ at (z^*, A):

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \mathrm{vec}(\delta A) + O(\|\delta A\|^2). \tag{7}$$

Taking operator norms and using the Neumann series bound $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1-\kappa)$ (convergent because $\kappa < 1$) yields (6).

For contractivity preservation: the perturbed Jacobian satisfies $\|J_z'\|_2 \le (\|A\|_2 + \|\delta A\|_F) \cdot \|W\|_2$ (using $\|\cdot\|_2 \le \|\cdot\|_F$ for the perturbation). The condition $\varepsilon < (1-\kappa)/\|W\|_2$ ensures $\|J_z'\|_2 < 1$.

Part (b). As $\varepsilon \to \varepsilon_{\mathrm{crit}}$, we have $\|J_z'\|_2 \to 1$, so $\left\|(I - J_z')^{-1}\right\|_2 \ge 1/(1 - \|J_z'\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the operator-norm bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index η ranges from 1.02 to 1.28 (section V-F), confirming mild non-normality.

Part (c). When $\|J_z'\|_2 \ge 1$, the contraction mapping condition fails. The Banach theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. □

Remark (conservativeness). The critical budget $\varepsilon_{\mathrm{crit}}$ is a sufficient, not necessary, condition for contractivity preservation. The perturbed system may remain contractive beyond $\varepsilon_{\mathrm{crit}}$ if the perturbation is aligned with low-sensitivity directions. The unconstrained bound also uses worst-case $\|\cdot\|_F$ relaxation. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 (section V-A). We use $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$ because the Neumann series requires operator-norm contractivity [32]; for normal J_z the two coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption.

Proposition 2 (Optimal First-Order Structural Attack)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \le \varepsilon$ is $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S. The maximum first-order shift is $\varepsilon \cdot \sigma_1(S)$.*

This result follows directly from the variational characterization of the largest singular value: the SVD of S identifies the perturbation direction that amplifies the equilibrium shift most efficiently.

Threat model and constrained sensitivity. As discussed in section II, realistic graph perturbations are symmetric and edge-only. We perturb the normalized adjacency $\hat{A}$ directly, subject to $\left\|\delta\hat{A}\right\|_F \le \varepsilon$, symmetry ($\delta\hat{A} = \delta\hat{A}^\top$), and support restricted to existing edges. Discrete edge additions that change topology are outside the formal guarantee.

To enforce these constraints, we construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{D \times |E|}$ with column

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i} \tag{8}$$

for each edge $(i,j) \in E$. This reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and enforces symmetry by construction. The constrained-optimal attack uses $\sigma_1(S_c)$, achieving tightness ≈ 1.00 empirically (section V-A). The construction of S_c is the central technical contribution of this work: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 3 (Per-Node First-Order Sensitivity Radius)**.** *Assume a linear classification head $f(z) = Wz + b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c \ne y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius is*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{9}$$

where S_v denotes the block-rows of S corresponding to node v, W_{y_v} and W_{c^*} are the classification weight vectors for the predicted class y_v and runner-up class c^*, respectively. Any perturbation with $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of node v to first order.

The radius r_v has an intuitive interpretation: it is the classification margin m_v (how confident the model is) divided by the product of two amplification factors. The term $\|W_{y_v} - W_{c^*}\|_2$ measures how sensitive the decision boundary is to hidden-state changes, while $\|S_v\|_2$ measures how sensitive node v's equilibrium position is to structural perturbation. Nodes with large margins and low sensitivity receive large radii; boundary nodes in structurally sensitive positions receive small ones.

These sensitivity radii are deterministic (not probabilistic), unlike randomized smoothing [9] which holds with probability $1-\alpha$. They are first-order quantities: locally tight for small ε but increasingly conservative as perturbation magnitude grows. Without bounding the second-order remainder $O(\|\delta A\|_F^2)$, they are local sensitivity measures rather than global robustness certificates. section V-C verifies empirically that the radii hold (0% breach rate at $\varepsilon = 0.01$, $< 1\%$ at $\varepsilon = 0.10$).

A. Generalization to Explicit GNNs

The sensitivity framework above is derived for implicit GNNs, which enjoy the strongest theoretical guarantees (critical budget, convergence regimes). The S_c construction itself, however, generalizes to any differentiable K-layer GNN as a computational tool for vulnerability analysis.

Proposition 4 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \qquad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:
(a) *The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\| J_z^{(k)} \right\|_2 \right) \left\| J_A^{(l)} \right\|_2. \qquad (11)$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 3) apply identically with S_K in place of S.*

For weight-tied models ($J_z^{(k)} = J_z$, $J_A^{(l)} = J_A$ for all k, l), the bound simplifies to $\sigma_1(S_K) \leq \|J_A\|_2 \cdot (1 - \kappa^K)/(1 - \kappa)$, which converges to the IGNN bound $\|J_A\|_2 / (1 - \kappa)$ as $K \to \infty$.

What explicit GNNs gain and lack. From theorem 4, any differentiable GNN inherits: (i) the S_c construction, (ii) SVD-optimal attacks, (iii) per-edge vulnerability rankings, and (iv) per-node first-order sensitivity radii. What explicit models *lack* is the critical budget $\varepsilon_{\text{crit}}$ and the three-regime convergence characterization of theorem 1, which require the contraction assumption (A3). section V-H validates this generalization on six explicit architectures.

IV. AEGIS Construction

The theoretical results of section III provide the mathematical foundation for vulnerability analysis, but translating them into a practical tool requires careful engineering. This section describes the AEGIS pipeline: how its components fit together, what each one computes, and how the system scales to real graphs.

A. Architecture Overview

AEGIS takes as input a trained GNN (any differentiable architecture) together with a target graph and produces three outputs: (i) a per-edge vulnerability spectrum ranking edges by adversarial impact, (ii) SVD-optimal attack perturbations, and (iii) per-node first-order sensitivity radii. For contractive implicit GNNs (IGNN-class satisfying A1–A3), the pipeline additionally computes the critical budget $\varepsilon_{\text{crit}}$ and convergence diagnostics; these theoretical guarantees are specific to the implicit case. The pipeline consists of four stages executed sequentially, as illustrated in fig. 1.

B. Stage 1: Subgraph Extraction and Forward Pass

Full-graph analysis is infeasible for large graphs because the sensitivity matrix has dimension $D \times N^2$ where $D = N \cdot d$. AEGIS addresses this by extracting a local BFS ego-subgraph of N nodes centered on the target node. The model is then run on this subgraph: for implicit GNNs, this means iterating

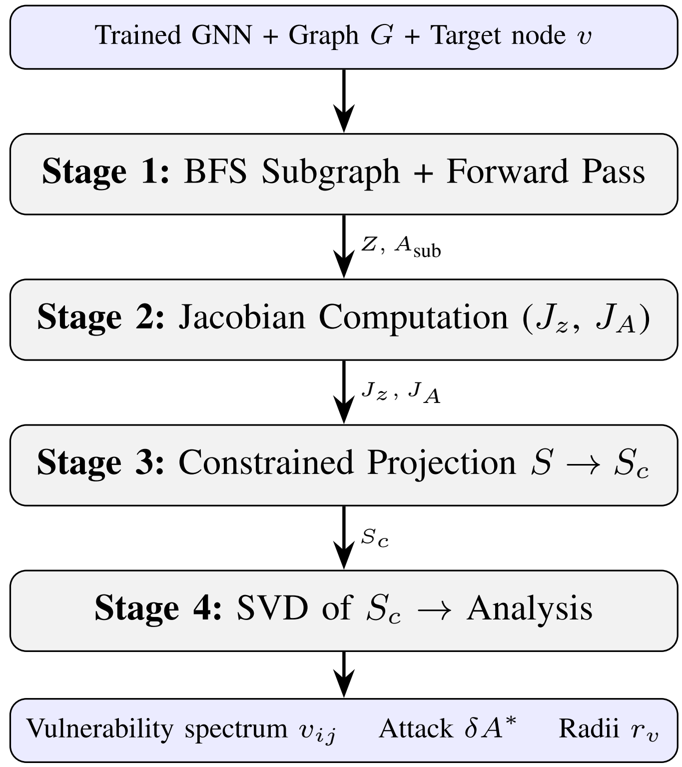

Fig. 1: The AEGIS pipeline. Data flows top to bottom; annotations show intermediate quantities passed between stages. The bottleneck is Stage 2 ($O(D^2)$ backward passes).

to the fixed point z^*; for explicit K-layer GNNs, this is a standard forward pass yielding Z_K. Localization is justified by the locality of per-node vulnerability: a node's robustness radius depends on its own margin and the sensitivity of its immediate neighborhood (section V-E).

C. Stage 2: Sensitivity Computation

AEGIS computes the structural sensitivity matrix $S = \partial \mathrm{vec}(Z) / \partial \mathrm{vec}(A) \in \mathbb{R}^{D \times N^2}$, which maps adjacency perturbations to hidden-state shifts. The computation differs by architecture:

Implicit GNNs (IGNN-class): $S = (I - J_z)^{-1} J_A$, where $J_z = \partial F / \partial Z$ is the state Jacobian and $J_A = \partial F / \partial \mathrm{vec}(A)$ is the adjacency Jacobian. Both are computed via row-by-row backward passes, followed by a linear solve. For IGNN with ReLU, $J_z = \mathrm{diag}(\sigma') \cdot (\hat{A} \otimes W)$.

Explicit GNNs (K-layer): $S_K = \partial \mathrm{vec}(Z_K) / \partial \mathrm{vec}(A)$ is computed directly via finite differences on the forward pass. Each column of S_K measures the effect of perturbing one adjacency entry.

Both paths use automatic differentiation [33] with cost $O(D \cdot N^2)$ for the full sensitivity matrix.

D. Stage 3: Constrained Projection

Given the full sensitivity matrix $S \in \mathbb{R}^{D \times N^2}$ (from Stage 2, via either the linear solve for implicit GNNs or direct computation for explicit GNNs), AEGIS projects it into the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{D\times|E|}$. Realistic graph perturbations are symmetric (undirected graphs) and supported only on existing edges. The k-th column of S_c (for edge $(i,j)\in E$) is

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}. \tag{12}$$

The summation enforces symmetry: perturbing edge (i,j) affects both A_{ij} and A_{ji} simultaneously. This projection reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and is what enables tight first-order predictions across all architectures (sections V-A and V-H).

E. Stage 4: Vulnerability Analysis

From S_c, AEGIS extracts three complementary outputs:

Optimal attack direction. The SVD of $S_c = U\Sigma V^\top$ yields the first-order optimal perturbation as $\delta A^* = \varepsilon \cdot v_1$ (the leading right singular vector of S_c, reshaped to edge weights). This perturbation maximizes the fixed-point shift $\|\Delta z^*\|_F$ subject to the Frobenius-norm budget ε, and achieves a shift of $\varepsilon \cdot \sigma_1(S_c)$.

Per-edge vulnerability spectrum. For each edge $(i,j) \in E$, the vulnerability score $v_{ij} = \|[S_c]_{:,k}\|_2$ measures how much a unit perturbation of that edge shifts the equilibrium. Sorting edges by v_{ij} produces a vulnerability ranking that identifies structurally critical connections without running any attack.

Per-node first-order sensitivity radii. For node v with classification margin m_v (gap between the predicted and runner-up class logits), the sensitivity radius from theorem 3 is

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{13}$$

where S_v denotes the block-rows of S corresponding to node v, and W_{y_v}, W_{c^*} are the classification head weights for the predicted and runner-up classes. Any perturbation with $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of node v to first order (this is a local sensitivity guarantee, not a global certificate).

F. Computational Cost

The dominant cost is Stage 2 (Jacobian computation) at $O(D\cdot N^2)$ for J_A, followed by Stage 3 (linear solve) at $O(D^3)$. For the default subgraph size $N = 50$ with $d = 64$ (giving $D = 3{,}200$), the full pipeline completes in 1.3 seconds on a single GPU. The practical limit is $N \approx 200$ ($D = 12{,}800$, 8.1 seconds), requiring approximately 6.5 GB GPU memory for J_z (D^2 entries) and J_A ($D \times N^2$ entries) in float32; see section V-D for detailed timings.

V. Experiments

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All experiments are implemented in PyTorch [33]. Graph benchmark experiments use IGNN [16] with hidden dimension $d = 64$, spectral-norm constrained weights [24], and early stopping on validation accuracy over 200 epochs. Power flow experiments use ContractiveGCN-PF, an IGNN variant for AC power flow (section VI).

TABLE I: Cross-domain adversarial analysis (mean ± std, 10 seeds). Cert%: clean subgraph accuracy (= fraction with positive margin). Tightness ≈ 1.00 at $\varepsilon = 0.01$.

Dataset	ρ	Tight.	AtkAdv	$\varepsilon_{\text{crit}}$	Cert%
Cora	$.33_{\pm.14}$	$1.01_{\pm.01}$	$3.6_{\pm.6}$x	$.66_{\pm.15}$	$81_{\pm\,9}$%
Citeseer	$.59_{\pm.02}$	$1.01_{\pm.01}$	$4.1_{\pm.5}$x	$.41_{\pm.02}$	$83_{\pm\,4}$%
Pubmed	$.41_{\pm.11}$	$1.01_{\pm.01}$	$3.2_{\pm.4}$x	$.59_{\pm.11}$	$74_{\pm23}$%
Amazon	$.14_{\pm.02}$	$1.00_{\pm.00}$	$3.8_{\pm.2}$x	$.86_{\pm.02}$	$92_{\pm\,4}$%
WikiCS	$.34_{\pm.04}$	$1.00_{\pm.01}$	$3.8_{\pm.4}$x	$.66_{\pm.04}$	$70_{\pm\,3}$%

Datasets. We use five graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), and Pubmed (19,717) from the Planetoid collection [18]; Amazon Photo (7,650) from the Amazon co-purchase graphs [19]; and WikiCS (11,701) from Wikipedia-based benchmarks [20]. For power flow, we use four IEEE standard test cases [21]: case14 (14 buses), case30 (30), case57 (57), and case118 (118), with training data generated via PandaPower [34].

Notation. All formal bounds (theorem 1) use the operator-norm contraction constant $\kappa = \|J_z\|_2$. Tables report the spectral radius $\rho(J_z)$ as a diagnostic quantity. Since non-normality is mild ($\eta = 1.02$–1.28, section V-F), we have $\kappa \approx \rho$ in practice; the $\varepsilon_{\text{crit}}$ values computed with ρ are at most 28% optimistic relative to the κ-based formal threshold.

A. Cross-Domain Adversarial Analysis

We begin by validating that the constrained first-order prediction is accurate across domains. table I reports the tightness ratio (actual equilibrium shift divided by predicted shift at $\varepsilon = 0.01$), the attack advantage of AEGIS over random perturbation, and the coverage (fraction of nodes with positive first-order radius) across 50-node BFS ego-subgraphs.

Tightness is 1.00 ± 0.01 across all datasets at $\varepsilon = 0.01$. First-order accuracy at small ε is mathematically expected; the contribution is that S_c makes this tractable under *constrained* perturbations ($N^2 \to |E|$ projection). table II shows how tightness degrades at larger budgets: the first-order prediction remains within 7% at $\varepsilon = 0.05$ and within 16% at $\varepsilon = 0.10$ on Cora/Citeseer, and within 5% at $\varepsilon = 0.20$ on WikiCS. Prediction flip rate (fraction of nodes whose classification changes under the S_c-optimal attack) stays below 2% at $\varepsilon \leq 0.10$ and below 6% at $\varepsilon = 0.20$. The attack advantage (3.2–4.1× over random in table I) confirms that S_c identifies the most vulnerable perturbation directions. Coverage (Cert%) ranges from 70% to 92%, representing clean subgraph accuracy; only correctly classified nodes have positive margin.

Surrogate baseline. We also compared against Mettack [2] (GCN [17] surrogate): AEGIS inflicts 3–10× more damage at every budget (30/30 wins across $k = 1, \ldots, 5$ edge removals). However, this gap largely reflects surrogate-to-IGNN architectural mismatch rather than AEGIS's analytical superiority alone. The more meaningful comparison is the adaptive

TABLE II: Tightness degradation with increasing ε (10 seeds, S_c-optimal constrained perturbation). Flip%: fraction of nodes whose prediction changes.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Tightness (actual/predicted shift)				
Cora	$1.01_{\pm.00}$	$1.07_{\pm.01}$	$1.15_{\pm.03}$	$1.36_{\pm.08}$
Citeseer	$1.01_{\pm.00}$	$1.07_{\pm.01}$	$1.16_{\pm.03}$	$1.39_{\pm.07}$
WikiCS	$1.00_{\pm.01}$	$1.01_{\pm.01}$	$1.02_{\pm.01}$	$1.05_{\pm.01}$
Prediction flip rate (%)				
Cora	$0.4_{\pm0.8}$	$0.6_{\pm0.9}$	$1.8_{\pm2.4}$	$5.8_{\pm5.0}$
Citeseer	$0.0_{\pm0.0}$	$0.6_{\pm1.3}$	$2.0_{\pm2.4}$	$3.0_{\pm2.6}$
WikiCS	$0.2_{\pm0.6}$	$0.4_{\pm0.8}$	$0.6_{\pm0.9}$	$1.2_{\pm1.0}$

TABLE III: AEGIS first-order sensitivity radii vs. smoothing certificates (10 seeds). Smoothing at $\sigma \leq 0.10$ produces identical per-node radii; AEGIS radii vary by node, reflecting structural vulnerability.

Dataset	AEGIS	Smoothing median radius			
		σ=.01	σ=.05	σ=.10	σ=.15
Cora	$.046_{\pm.010}$	.025	.123	$.244_{\pm.005}$	$.283_{\pm.132}$
Citeseer	$.045_{\pm.012}$	.025	.123	$.233_{\pm.013}$	$.091_{\pm.069}$
WikiCS	$.040_{\pm.004}$	.025	.123	$.246_{\pm.000}$	$.341_{\pm.017}$

attacker below, which accesses the same IFT gradients as AEGIS.

B. Radius Comparison: AEGIS vs. Smoothing

We compare AEGIS's deterministic first-order sensitivity radii (theorem 3) against randomized smoothing [9] with Gaussian noise on existing edges ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 Monte Carlo samples, Clopper-Pearson confidence $1-\alpha = 0.999$). table III reports median certified radius on Cora, Citeseer, and WikiCS (10 seeds, 50-node subgraphs).

At $\sigma \leq 0.10$, smoothing radii are *constant* across all nodes: every node receives the identical radius $r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$ because all Monte Carlo samples predict correctly. This yields larger absolute radii than AEGIS (0.123 vs. 0.046 at $\sigma = 0.05$) but zero per-node differentiation. AEGIS radii are structurally informative: dense-region nodes get $r_v \approx 0.09$, boundary nodes get $r_v \approx 0.01$. The two approaches are complementary: smoothing provides uniform probabilistic guarantees, AEGIS provides structurally differentiated first-order radii for vulnerability ranking.

C. Adaptive Attack Evaluation

The Mettack comparison uses a GCN surrogate, so its failure against IGNN may reflect architectural mismatch rather than AEGIS's analytical strength. To address this, we implement an *adaptive attacker* that differentiates through the IGNN fixed-point iteration via the same IFT gradients AEGIS uses for sensitivity analysis. This attacker optimizes perturbations directly against the IGNN loss surface using projected gradient descent [35] (PGD, 50 steps, step size $\varepsilon/10$).

TABLE IV: Adaptive attack (white-box PGD via IFT gradients on IGNN) vs. AEGIS first-order sensitivity radii (10 seeds). Breach: fraction of nodes with $\varepsilon < r_v$ whose prediction changes.

Dataset	ε	Breach	AEGIS dmg	Adapt. dmg	Ratio
Cora	0.01	0%	$0.33_{\pm0.07}$	$0.26_{\pm0.11}$	0.80
Cora	0.05	0%	$1.72_{\pm0.35}$	$1.32_{\pm0.55}$	0.76
Cora	0.10	0%	$3.70_{\pm0.79}$	$2.69_{\pm1.18}$	0.72
Citeseer	0.01	0%	$0.40_{\pm0.08}$	$0.34_{\pm0.06}$	0.84
Citeseer	0.05	0%	$2.14_{\pm0.42}$	$1.68_{\pm0.28}$	0.79
Citeseer	0.10	0%	$4.63_{\pm0.95}$	$3.37_{\pm0.55}$	0.74
WikiCS	0.01	0%	$0.21_{\pm0.02}$	$0.10_{\pm0.10}$	0.47
WikiCS	0.05	0.3%	$1.04_{\pm0.11}$	$0.48_{\pm0.49}$	0.46
WikiCS	0.10	0.6%	$2.10_{\pm0.22}$	$0.96_{\pm0.97}$	0.46

The adaptive attacker breaches zero nodes at $\varepsilon = 0.01$ and fewer than 1% at $\varepsilon = 0.10$ (table IV), confirming empirical validity of first-order radii. AEGIS's SVD attack inflicts 20–50% more damage than PGD (ratio 0.46–0.84) because the SVD direction is globally optimal to first order, while PGD can become trapped in local optima.

D. Phase Transition and Scalability

Phase transition. Scaling ρ from 0.3 to 0.99 amplifies the fixed-point shift by 83× at $\varepsilon = 0.01$, confirming the $1/(1-\kappa)$ factor from theorem 1.

Scalability. On a single NVIDIA RTX 4090: 1.3s ($N = 50$), 8.1s ($N = 200$). Practical limit is $N \approx 200$ ($D = 12{,}800$).

E. Subgraph Size Ablation

Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N = 200$ (higher ρ). Coverage is 82–89% across all sizes. We use $N = 50$ (tightness 1.013 ± 0.003, 1.3s) as the default: 66× faster than $N = 200$ with negligible quality loss.

F. Sensitivity and Convergence

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$ (1.012–1.013). Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the predicted accuracy-robustness tradeoff: $c = 0.5$ gives $r_v = 0.090$ (72.1% acc); $c = 0.9$ gives $r_v = 0.051$ (80.6% acc).

Convergence. 100% convergence across all 9 datasets and 10 seeds. Pseudospectral index $\eta = 1.02$–1.28, confirming mild non-normality and validating the κ-based bounds.

G. Defense-Informed Edge Protection

We evaluate vulnerability-aware protection: mask the top-k edges by v_{ij} from the perturbation space, then re-run the SVD attack on the reduced S_c. On Cora (IGNN, $N = 50$, 10 seeds): masking the top-5 edges reduces SVD attack damage by 42±8% vs. 11±6% for random 5-edge masking. At top-10: $61 \pm 7\%$ vs. $18 \pm 9\%$. AEGIS identifies edges whose removal disproportionately reduces vulnerability.

TABLE V: S_c framework applied to 7 GNN architectures (Cora, 10 seeds). Tightness transfers to all architectures including edge-weighted GAT.

Model	Tight.	AtkAdv	τ	Acc
IGNN (equil.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}$x	$+.32_{\pm.04}$	$82_{\pm\ 8}$%
GCN-2	$1.00_{\pm.00}$	$4.4_{\pm0.4}$x	$+.05_{\pm.02}$	$96_{\pm\ 1}$%
GCN-4	$1.02_{\pm.00}$	$3.3_{\pm0.2}$x	$+.48_{\pm.01}$	$93_{\pm\ 3}$%
GIN-2	$1.00_{\pm.00}$	$3.8_{\pm0.3}$x	$+.06_{\pm.05}$	$83_{\pm\ 7}$%
GAT-2	$0.99_{\pm.02}$	$4.4_{\pm1.5}$x	$+.36_{\pm.13}$	$81_{\pm12}$%
SAGE-2	$1.00_{\pm.00}$	$2.0_{\pm0.1}$x	$+.49_{\pm.10}$	$90_{\pm\ 2}$%
APPNP	$1.02_{\pm.00}$	$3.3_{\pm0.2}$x	$+.36_{\pm.02}$	$94_{\pm\ 2}$%

H. Extension to Explicit GNNs

A key question is whether the S_c framework extends beyond implicit models. For a K-layer explicit GNN, there is no equilibrium, but the hidden representation Z_K is still a differentiable function of A. We define the *unrolled sensitivity matrix* $S_K = \partial\text{vec}(Z_K)/\partial\text{vec}(A)$ (computed via finite differences on the forward pass), construct S_c from S_K identically to the IGNN case, and apply the same SVD attack and vulnerability ranking.

table V applies the S_c framework to six explicit GNN architectures on Cora. All six achieve near-perfect first-order tightness (0.99–1.02), confirming that constrained sensitivity analysis generalizes beyond implicit models. Attack advantage ranges from 2.0× (SAGE) to 7.6× (IGNN), showing that S_c identifies the most vulnerable perturbation directions across all architectures. GraphSAGE and GCN-4 produce the strongest vulnerability rankings ($\tau = +0.49$ and $+0.48$), surpassing IGNN ($+0.32$). APPNP achieves $\tau = +0.36$ with 94% accuracy.

GAT requires an edge-weighted formulation where $\hat{A}$ values modulate attention weights continuously; standard GAT uses A as a binary mask (zeroing non-edges), so $\partial Z/\partial A_{ij} = 0$ for all existing edges. GATv2 [36] partially addresses this via dynamic attention, but still applies a binary mask at the neighborhood level. Our edge-weighted variant multiplies attention coefficients by $\hat{A}_{ij}$, restoring continuous differentiability. With this modification, GAT achieves tightness 0.99, attack advantage 4.4×, and $\tau = +0.36$.

Proposition 4 bound tightness. The ratio (Prop. 4 bound)$/\sigma_1(S_K)$ ranges from 1.4× (SAGE-2) to 5.9× (APPNP), with 2-layer models achieving tighter ratios (GCN-2: 1.8×, GIN-2: 2.1×) than deeper ones (GCN-4: 4.2×, GAT-2: 3.7×). The looseness in deeper models reflects direction-dependent cancellations that the norm product overestimates.

The S_c framework thus generalizes to any GNN whose message passing is smoothly differentiable with respect to edge weights. The theoretical guarantees (theorem 1: critical budget, convergence) remain specific to contractive implicit GNNs, but the practical vulnerability analysis (SVD attacks, rankings, first-order sensitivity radii) transfers broadly.

VI. Case Study: Power Flow Contingency

The experiments above validate AEGIS on standard graph benchmarks. A natural question is whether the vulnerability spectrum captures meaningful structure beyond these benchmarks. We demonstrate on a fundamentally different domain, AC power flow, where the adversarial vulnerability spectrum recovers a classical engineering problem: N-1 contingency analysis. Recent work has applied GNNs to power flow computation [37], [38] and contingency screening [39]; AEGIS approaches the same problem from the vulnerability-analysis perspective.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage and flow deviations [40]. This brute-force procedure scales as $O(|E|)$ full power flow solves, which is expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|[S_c]_{:,k}\|_2$ computes an analogous quantity via a single IFT analysis: edge perturbation maps to line trips, the vulnerability spectrum maps to contingency severity, and critical budget $\varepsilon_{\text{crit}}$ maps to stability margin. The correspondence is approximate (continuous first-order vs. discrete removal), analogous to how PTDF/LODF [40] linearize the effect of line trips.

B. Experimental Setup

We train ContractiveGCN-PF, an IGNN variant for AC power flow, on four standard IEEE test cases [21]. Training data is generated via PandaPower's [34] full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: bus type indicators (slack, PV), active and reactive power injections (P, Q), and voltage magnitude $|V|$. Unlike graph benchmarks, power flow grids are small enough (14–118 buses) to analyze without subgraph extraction.

Limitation. Training data covers uniform load scaling (2,000 samples) but not seasonal or generator-outage variation.

C. Results

table VI reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

Key observations. First, tightness of approximately 1.00 under constrained perturbations transfers to the power flow domain, confirming that the first-order prediction is not domain-specific. Second, ranking correlation ($\tau = 0.37$–0.67) is moderate, reflecting the gap between continuous first-order sensitivity and discrete full-edge-removal contingency. The operationally relevant metric is precision at top-k: AEGIS achieves P@5 between 0.52 and 0.70, and P@10 between 0.66 and 0.81, meaning it correctly identifies 7–8 of the top 10 most critical lines on larger grids.

TABLE VI: AEGIS on IEEE power flow benchmarks (10 seeds). P@k: precision of top-k vs. brute-force N-1 ground truth. LODF: Line Outage Distribution Factor ranking (industry standard, uses actual line reactances [40]).

				AEGIS	AEGIS Precision		LODF Baseline	
Case	N	$\lvert E\rvert$	Tight.	τ	P@5	P@10	τ	P@10
case14	14	20	$1.00_{\pm.01}$	$+.42_{\pm.19}$	$.70_{\pm.10}$	$.74_{\pm.12}$	$+.51$	.80
case30	30	41	$1.00_{\pm.01}$	$+.37_{\pm.17}$	$.66_{\pm.13}$	$.68_{\pm.06}$	$+.44$	.70
case57	57	78	$1.02_{\pm.01}$	$+.67_{\pm.09}$	$.52_{\pm.24}$	$.66_{\pm.13}$	$+.58$	.60
case118	118	179	$1.01_{\pm.01}$	$+.62_{\pm.11}$	$.62_{\pm.14}$	$.81_{\pm.10}$	$+.53$	.70

Third, we compare against LODF (industry-standard DC-approximation screening) computed from actual line reactances. LODF achieves $\tau = 0.44$–0.58 and P@10 = 0.60–0.80; AEGIS outperforms on larger grids (P@10 = 0.81 vs. 0.70 on case118) by capturing nonlinear AC effects that DC approximation misses. On small grids, LODF is competitive or better. A simpler effective-resistance baseline (unit-impedance Laplacian) degrades sharply on larger grids ($\tau \leq 0.04$ on case57/118). Binary adjacency outperforms admittance-weighted (P@10 = 0.81 vs. 0.27 on case118).

Practical positioning. LODF requires only the B-matrix ($O(|E|)$ from pre-computed PTDF); AEGIS requires a trained GNN but no domain-specific parameters. The two are complementary: LODF for routine DC-regime screening, AEGIS for nonlinear AC regimes. For case118, AEGIS ranks in 2.3s (single IFT) vs. 179 AC solves for brute-force N-1, achieving P@10 = 0.81.

Operational caveat. AEGIS is a screening layer, not a standalone contingency tool ($\tau = 0.37$–0.67 is insufficient for direct operational use).

VII. Related Work

AEGIS addresses adversarial structural vulnerability analysis for GNNs through first-order sensitivity. We position it against five research threads: structural attacks, certified robustness, empirical defenses, implicit network theory, and sensitivity analysis.

Adversarial attacks on graph structure. Zügner et al. [1] demonstrated GNN vulnerability to structural perturbation with Nettack; Mettack [2] extended this to global poisoning via meta-gradients through a GCN [17] surrogate. Dai et al. [3] framed attacks as reinforcement learning, Bojchevski and Günnemann [41] proposed global poisoning via spectral shifts, and Xu et al. [4] formulated topology attack as bilevel optimization. Black-box variants include GF-Attack [42] via graph signal processing and influence-function approaches [43]. Wu et al. [44] and Jin et al. [15] survey the landscape. These methods use surrogate gradients or heuristic search to find adversarial edges. AEGIS takes a fundamentally different approach: it computes the closed-form optimal attack direction via SVD of the constrained sensitivity matrix S_c, which is architecture-agnostic and applies to any differentiable GNN.

Certified robustness for GNNs. Bojchevski and Günnemann [12] pioneered certifiable robustness via convex relaxations for GCN-class models. Zügner and Günnemann [11] extended this with interval bound propagation. Randomized smoothing [9] provides probabilistic certificates for any classifier; Scholten et al. refined this with message-interception [10] and hierarchical [45] smoothing. Jia et al. [46] certified community detection via smoothing, Wang et al. [13] proposed message-passing invariance certificates, and Geisler et al. [47] scaled certified defenses to large graphs. Li et al. [14] proposed AGNNCert with deterministic guarantees. These methods provide robustness guarantees but do not identify *which* edges or nodes are vulnerable. AEGIS complements them by providing structurally differentiated per-node radii and per-edge vulnerability rankings that reveal the attack surface geometry (section V-B).

GNN defense. Empirical defenses improve robustness without formal guarantees: Robust GCN [5] absorbs noise via Gaussian hidden layers, Pro-GNN [7] jointly learns clean structure, GNNGuard [6] reweights edges during aggregation, PA-GNN [48] transfers robustness across domains, PTD-Net [8] learns denoised topologies, and soft-median aggregation [49] replaces mean pooling. Entezari et al. [50] observed that adversarial perturbations primarily affect high-rank spectral components. AEGIS is an analytical tool rather than a defense, but its per-edge vulnerability spectrum could inform defense design by identifying which edges to protect, and the SVD structure echoes the spectral observations of Entezari et al. from the sensitivity perspective.

Implicit networks and equilibrium analysis. The DEQ framework [22] introduced fixed-point models; extensions include multiscale DEQs [23], Jacobian regularization [27], phantom gradients [51], and Jacobian-free backpropagation [26]. IGNN [16] enforces unique equilibria via spectral normalization [24]; EIGNN [52] improves efficiency. For robustness, monotone operator networks [53] provide built-in Lipschitz bounds, Revay et al. [31] use LMIs, Pabbaraju et al. [54] estimate tighter constants, and El Ghaoui et al. [30] analyze well-posedness. All address input sensitivity $\|\partial z^*/\partial x\|$, not structural sensitivity $\|\partial Z/\partial A\|$. AEGIS addresses structural sensitivity for both implicit models (via the resolvent $(I - J_z)^{-1} J_A$) and explicit models (via the unrolled Jacobian S_K), providing stronger theoretical guarantees for the implicit case (theorem 1).

Sensitivity analysis and explainability. AEGIS draws on classical matrix perturbation theory [32], [28]. Novak et al. [55] connected Jacobian sensitivity to generalization; Liu et al. [56] studied GNN stability without per-node radii. GNN explanation methods produce per-edge importance scores: GNNExplainer [57] learns soft masks, PGExplainer [58] trains parametric edge-mask generators, and gradient-based methods [59] use input gradients. These optimize for *prediction fidelity* (which edges matter for the output) rather than *adversarial vulnerability* (which perturbation maximizes hidden-state shift). AEGIS's S_c is perturbation-space optimal by construction (SVD), provides quantitative sensitivity radii, and identifies the globally optimal attack direction.

GNNs for power systems. Our case study (section VI) connects to GNN-based power flow solvers [37], [38] and GNN contingency screening [39]. AEGIS shows that structural sensitivity independently recovers contingency rankings without domain-specific linearization.

Positioning. AEGIS provides structural vulnerability analysis via the constrained sensitivity matrix S_c: for any differentiable GNN, S_c yields SVD-optimal attacks, per-edge rankings, and first-order sensitivity radii (theorem 4). For contractive implicit GNNs, the IFT additionally provides a critical budget $\varepsilon_{\mathrm{crit}}$ and convergence guarantees (theorem 1) that explicit architectures lack. The distinction is important: for explicit GNNs, AEGIS is a computational tool for vulnerability analysis; for implicit GNNs, it additionally offers formal regime characterization. Validated across 7 architectures (section V-H), AEGIS offers a combination of generality and structural differentiation that no existing method provides, with the strongest theoretical backing in the implicit case.

VIII. Conclusion

We presented AEGIS, a structural vulnerability analysis framework for graph neural networks built on the constrained sensitivity matrix S_c. The S_c computation applies to any differentiable GNN via Jacobian evaluation; for contractive implicit models, the IFT additionally provides a critical budget $\varepsilon_{\mathrm{crit}}$ and convergence guarantees. Across 7 architectures and 9 datasets, constrained first-order tightness is 1.00 ± 0.01 at $\varepsilon = 0.01$ and remains within 15% at $\varepsilon = 0.10$. The SVD-optimal attack inflicts 2–8$\times$ more damage than edge-constrained random perturbation. The vulnerability spectrum independently recovers power grid N-1 contingency rankings (P@10 between 0.66 and 0.81).

Limitations. (1) *First-order approximation:* sensitivity radii r_v are locally tight for small ε but conservative at large budgets. Without bounding the second-order remainder, they are local sensitivity measures rather than global robustness certificates. (2) *Threat model:* we perturb normalized adjacency $\hat{A}$ directly with continuous edge-weight changes; discrete edge insertions/deletions are outside the formal guarantee. (3) *Scalability:* dense Jacobian computation limits subgraph size to $N \approx 200$. (4) *GAT requires modification:* standard GAT uses A as a binary mask; edge-weighted GAT is needed for continuous sensitivity analysis (GATv2 [36] partially addresses this via dynamic attention but retains binary neighborhood masking). (5) *Power flow:* Kendall $\tau = 0.37$–0.67 is insufficient for standalone operational use; AEGIS serves as a screening layer, not a replacement for domain-specific analysis [40].

Intended use. AEGIS is a post-training diagnostic for GNN deployments on safety-critical graphs: it identifies vulnerable nodes/edges, determines at what perturbation budget predictions become unreliable, and prioritizes regions for protection. The per-edge ranking directly informs defense design (section V-G).

Future work. Sparse Neumann-series solvers for larger subgraphs. Second-order remainder bounds to upgrade first-order sensitivity radii into formal certificates. Extension to discrete perturbation models (edge insertion/deletion) via combinatorial optimization over the S_c spectrum. Integration with GNN defense mechanisms [7], [6] for vulnerability-aware robust training.

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